

MATH GUESS

Class 11 Federal Board

Chapter # 01

Example 8: Find the values of x and y if, $\frac{x}{2+3i} - y(1+2i) = 1+i$

5. Convert the following equations and inequations in Cartesian form:

$$(iii) -\frac{\pi}{3} \leq \arg(z-4) \leq \frac{\pi}{3}$$

$$(v) \arg\left(\frac{1-iz}{1-z}\right) = \frac{\pi}{4}; z \neq i$$

$$(vi) \frac{1}{2}\arg(z-i) = \frac{\pi}{3} - \frac{1}{2}\arg(z+i)$$

Example 20: Find the principal arg z , when $z = \frac{-2}{1+i\sqrt{3}}$.

Example 24: Find the equation in Cartesian form, if $z = x + iy$ and $\arg(z-2) - \arg(z+2) = \frac{\pi}{4}$.

Example 16: Factorize the polynomial $P(z) = z^3 + 2z^2 - 5z - 6$.

4. Solve the simultaneous system of linear equations with complex coefficients:

$$(iii) \quad \frac{3}{i}z - (6 + 2i)\omega = 5; \quad \frac{i}{2}z + \left(\frac{3}{4} - \frac{1}{2}i\right)\omega = \left(\frac{1}{2} + 2i\right)$$

$$(iv) \quad \frac{1}{1-i}z + (1 + i)\omega = 3; \quad \frac{2}{i}z - (2 - 3i)\omega = 2 + 6i$$

Example 12: If $x_1 + iy_1 = 12 + 5i$ and $x_2 + iy_2 = 3 + 2i$ then find the real and imaginary parts of the following:

$$(ii) \quad \left(\frac{x_1 + iy_1}{x_2 + iy_2}\right)^{-1} \quad (iv) \quad \left(\frac{x_1 + iy_1}{x_2 + iy_2}\right)^{-2}$$

3. Solve the following quadratic equations:

$$(i) \quad \frac{1}{3}z^2 + 2z - 16 = 0 \quad (ii) \quad z^2 - \frac{1}{2}z + 17 = 0$$

Example 15: Solve the following system of simultaneous equations:

$$2z - (1 - 3i)\omega = 1 + 2i, \quad (1 + i)z + (2 - i)\omega = 2 + i$$

Example 13: Write the equation $|z - 2i| = |\bar{z} + 3|$ in terms of x and y , by taking $z = x + iy$.

Example 17: Solve the equation $2z^2 - 6z - 9 = 0$ by completing square method.

Chapter # 02

Example 9: Write the matrix $A = \begin{bmatrix} 3 & 1 & 5 \\ 2 & 6 & 0 \\ -1 & 2 & 1 \end{bmatrix}$ as a sum of two matrices where one is symmetric and the other is skew-symmetric.

Example 22: Without expansion show that $\begin{vmatrix} a-2l & b-2m & c-2n \\ l & m & n \\ a & b & c \end{vmatrix} = 0$
Solution: L.H.S

Example 23: If $a + b + c = 0$; then without expanding show that

$$\begin{vmatrix} b+c & c+a & a+b \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{vmatrix} = 0$$

Example 24: Prove that $\begin{vmatrix} a^2+1 & ab & ac \\ ab & b^2+1 & bc \\ ac & bc & c^2+1 \end{vmatrix} = 1 + a^2 + b^2 + c^2$

Example 25: Prove that $\begin{vmatrix} 1 & a^2 & a \\ a & 1 & a^2 \\ a^2 & a & 1 \end{vmatrix} = (1 - a^3)^2$

1. Without expansion show that:

(iii) $\begin{vmatrix} 0 & -a & -b \\ a & 0 & -c \\ b & c & 0 \end{vmatrix} = 0$

$$(ii) \begin{vmatrix} 1/a & bc & b+c \\ 1/b & ac & a+c \\ 1/c & ab & a+b \end{vmatrix} = 0$$

2. Using the properties of the determinants prove the following.

$$(i) \begin{vmatrix} x & y & x+y \\ y & x+y & x \\ x+y & x & y \end{vmatrix} = -2(x^3 + y^3)$$

$$(ii) \begin{vmatrix} a & b-c & b+c \\ a+c & b & c-a \\ a-b & a+b & c \end{vmatrix} = (a+b+c)(a^2 + b^2 + c^2)$$

$$(iv) \begin{vmatrix} x & x^2 & 1+\alpha x^3 \\ y & y^2 & 1+\alpha y^3 \\ z & z^2 & 1+\alpha z^3 \end{vmatrix} = (1+\alpha xyz)(x-y)(y-z)(z-x)$$

$$(vi) \begin{vmatrix} 3a & 1 & 2a+1 \\ 2a+1 & 1 & a+2 \\ 3 & 1 & 2 \end{vmatrix} = (a-1)(a-2)$$

$$(vii) \begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = 4abc$$

Example 28: Find the rank of the matrix $\begin{bmatrix} 2 & 5 & 7 \\ 1 & 2 & -1 \\ -3 & -6 & 3 \end{bmatrix}$.

Example 29: Find A^{-1} , if $A = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 3 & 1 \\ 1 & 0 & 2 \end{bmatrix}$ by using row operations.

2. Find the rank of each of the following matrices.

$$(ii) \begin{bmatrix} -1 & -2 & 3 \\ -1 & 2 & -1 \\ -5 & 2 & 3 \end{bmatrix}$$

$$(iii) \begin{bmatrix} 3 & 2 & 4 \\ 2 & 1 & 6 \\ 4 & -1 & 0 \end{bmatrix}$$

Example 31: Solve the homogeneous system of linear equations

$$x + 3y + 2z = 0$$

$$2x - y + 3z = 0$$

$$x - 4y + z = 0$$

Example 32: Solve the system of non-homogeneous linear equation by matrix inversion method.

$$2x + 3y - z = 1; \quad x - y + z = 3; \quad x + 2y - z = 1$$

Example 33: Solve the system of equations by using Gauss elimination method

$$2x_1 - 3x_2 + 4x_3 = 1; \quad x_1 + 2x_2 - x_3 = 2; \quad 3x_1 + 5x_2 - 3x_3 = 5$$

Solution:

Example 34: Solve the system of given non-homogeneous linear equations

$$2x - 3y + 5z = 2; \quad x + 4y - 2z = 1; \quad 4x + 5y + z = 4,$$

Solution:

Example 35: Solve the given system of non-homogeneous linear equations

$$2x - 3y + 5z = 1; \quad x + y + 2z = 3; \quad 3x - 2y - 4z = 0$$

by Cramer's rule.

Determine the value of λ for which the following system has no solution, unique solution or infinitely many solutions.

$$x + 2y - 3z = 4; \quad 3x - y + 5z = 2; \quad 4x + y + (\lambda^2 - 14)z = \lambda + 2$$

5. Solve the following system of linear equations by matrix inversion method.

$$\begin{aligned} \text{(ii)} \quad & x + 2y - 3z = 5 \\ & 2x - 3y + 2z = 1 \\ & -x + 2y - 5z = -3 \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad & -x + 3y - 5z = 0 \\ & 2x + 4y - 6z = 1 \\ & x - 2y + 3z = 3 \end{aligned}$$

2. Find the value of λ for which the following system of homogeneous linear equations may have non-trivial solution. Also solve the system for value of λ .

$$\begin{aligned} \text{(i)} \quad & 2x_1 - \lambda x_2 + x_3 = 0 \\ & 2x_1 + 3x_2 - x_3 = 0 \\ & 3x_1 - 2x_2 + 4x_3 = 0 \end{aligned}$$

Chapter # 03

Example 4:

Find the vector which divides $\overrightarrow{OA} = \underline{a} = 3\hat{i} + \hat{j} + 10\hat{k}$ and $\overrightarrow{OB} = \underline{b} = 3\hat{i} + 7\hat{j} + \hat{k}$ in the ratio 3:2(Internally)

Example 5:

Find the vector which divides $\overrightarrow{OA} = \underline{a} = 3\hat{i} + \hat{j} + 10\hat{k}$ and $\overrightarrow{OB} = \underline{b} = 3\hat{i} + 7\hat{j} + \hat{k}$ in the ratio 3:2(externally)

8. If $\vec{a} = 2\hat{i} - 3\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - 3\hat{j} + 5\hat{k}$ then find:
- (i) A vector of magnitude 5 in the direction $\vec{a} - 2\vec{b}$.
 - (ii) A vector of magnitude $\frac{3}{7}$ opposite in direction $3\vec{a} + \vec{b}$.
9. The position vectors of points A and B are $\hat{i} - 2\hat{j} + \hat{k}$ and $2\hat{i} + 3\hat{j} - \hat{k}$ respectively.
- (i) Find the position vector of point P dividing the line segment joining A and B in the ratio 2 : 3 internally.
 - (ii) Find the position vector of point Q dividing the line segment $\overline{AB}$ in the ratio 3 : 2 externally.
10. (i) The three vertices of a parallelogram ABCD taken in order are $A(3, -4)$; $B(-1, -3)$ and $C(-6, 2)$. Find the fourth vertex D.

Example 9: Find the coordinates of point P, if $\overrightarrow{OP}$ is a vector of magnitude 2 and is parallel to the vector $2\hat{i} - 3\hat{j} + 4\hat{k}$.

Example 10: Two direction angles of vector $\vec{r}$ are 30° and 60° . Find the third direction angle. Also find unit vector $\hat{r}$.

Example 11: Find the angle between the vectors $\hat{i} - 2\hat{j} + \hat{k}$ and $2\hat{i} - 3\hat{j} + \hat{k}$.

Example 12: If $\vec{a} = \hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = -\hat{i} + \hat{j} + 3\hat{k}$, find projection of $\vec{a}$ along $\vec{b}$ and projection of $\vec{b}$ along $\vec{a}$.

Example 13: Prove that $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$

Example 14: For any triangle ABC, with usual notations prove that $|\vec{a}| = |\vec{b}| \cos \gamma + |\vec{c}| \cos \beta$

Example 15: Find the work done by a constant force $\vec{F} = 2\hat{i} + \hat{j} - \hat{k}$ in moving an object from $A(0,1,3)$ to $B(-1,2,4)$.

- (ii) If $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$, then find angle between $\vec{a}$ and $\vec{b}$.

4. (i) If $\vec{a} = \hat{i} - 3\hat{j} + 4\hat{k}$; $\vec{b} = 7\hat{i} - 9\hat{j} + \hat{k}$ and $\vec{c} = 3\hat{i} - 2\hat{j} + 5\hat{k}$, find the value of λ so that $\vec{a} - \lambda\vec{b}$ is perpendicular to $\vec{c}$.
- (ii) Show that the angle between any two diagonals of a cube is $\cos^{-1}\left(\frac{1}{3}\right)$.
- (ii) If $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$; $\vec{b} = 3\hat{i} - 2\hat{j} + \hat{k}$ and $\vec{c} = 7\hat{i} - \hat{j} + 8\hat{k}$, then find the projection of $\vec{a} - \vec{b}$ along $\vec{c}$ and projection of $\vec{b}$ along $\vec{c} - \vec{a}$. Also find their vector projections.
11. (i) Prove that altitudes of a triangle are concurrent.
- (ii) Prove that angle bisectors of triangle are concurrent.
12. (i) Prove that $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$.
- (ii) With usual notations for a triangle ABC; prove that $c^2 = a^2 + b^2 - 2ab \cos \gamma$ and $b = a \cos \gamma + c \cos \alpha$
13. (i) The resultant of two vectors $\vec{a}$ and $\vec{b}$ is perpendicular to $\vec{a}$ and $|\vec{a}| = \frac{1}{\sqrt{7}}|\vec{b}|$. Show that resultant of $7\vec{a}$ and $\vec{b}$ is perpendicular to $\vec{b}$.
- (ii) Prove that $\vec{a} + \vec{b}$ is equally inclined with $\vec{a}$ and $\vec{b}$.
14. A force of $\vec{F} = 3\hat{i} - 5\hat{j} + 7\hat{k}$ newtons is applied on a body and moves it at a distance of 14 meters in the direction of the vector $\hat{i} - 3\hat{j} + \hat{k}$. How much work is done?
15. Find the work done by the forces $2\hat{i} + 3\hat{j} - \hat{k}$ and $3\hat{i} + 7\hat{j} + 4\hat{k}$ acting on a particle in moving from point P to Q with position vectors $\hat{i} - 3\hat{j} + \frac{3}{2}\hat{k}$ and $2\hat{i} - \hat{j} + \frac{5}{2}\hat{k}$.

Example 17: Find the area of a triangle with vertices $(0, 0)$, $(2, 9)$, $(3, 5)$.

Example 18: If $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$; $\vec{b} = -\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{c} = \hat{i} + \hat{j} - 3\hat{k}$ then find the angle between the vectors $\vec{a} + \vec{b}$ and $\vec{a} + \vec{c}$.

Example 19: Show that $\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$.

Example 20: Find the moment of the force $\vec{F} = 3\hat{i} - 2\hat{j} + 5\hat{k}$ about the point $(2, 1, -1)$ when it is applied at point $(3, 0, 2)$.

Example 21: Find the moment of the force $\vec{F} = 7\hat{i} + 4\hat{j} + 2\hat{k}$ when it is applied at the handle of a door at the point $(2, 1, 4)$ about the hinge at point $(0, 0, 1)$.

4. i. Find a vector of magnitude 5 and perpendicular to both the vectors $\vec{a} = 3\hat{i} - 2\hat{j} + 5\hat{k}$ and $\vec{b} = 8\hat{i} - 2\hat{j} + \hat{k}$.
- ii. Express the vector $5\hat{i} + 2\hat{j} - 3\hat{k}$ as a sum of two vectors one of which is parallel and other is perpendicular to the vector $2\hat{i} - \hat{j} + 3\hat{k}$.
5. i. Prove the Lagrange identity $|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2$

(ii) For three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ if $\vec{a} + \vec{b} + \vec{c} = 0$, then prove that :

$$\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$$

(ii) Prove that $|\vec{a} \times \vec{b}|^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} \\ \vec{a} \cdot \vec{b} & \vec{b} \cdot \vec{b} \end{vmatrix}$

9. (i) If $|\vec{a}| = 3$; $|\vec{b}| = 5$ and $\vec{a} \cdot \vec{b} = \frac{15}{4}$ then find $|\vec{a} \times \vec{b}|$.

(ii) If $|\vec{a}| = 2$; $|\vec{b}| = 5$ and $|\vec{a} \times \vec{b}| = 8$ then find $\vec{a} \cdot \vec{b}$.

14. (i) Find the moment of force $2\hat{i} + 3\hat{j} + 7\hat{k}$ about the point $(1, 2, 3)$ when applied at the point $(-1, 2, 0)$.

(ii) Two forces $2\hat{i} - \hat{j} + 3\hat{k}$ and $3\hat{i} + 4\hat{j} - 2\hat{k}$ are applied at the same point $(1, -2, 4)$. Find the moment of these concurrent forces about $(0, 0, 0)$.

15. (i) How much force is required to produce a moment of magnitude $\sqrt{57} \text{ N.m}$ along the direction $6\hat{i} - 21\hat{j} - 6\hat{k}$ when applied at $(2, 1, -3)$ about $(-1, -1, 1)$.

(ii) At what point the force $2\hat{i} + 2\hat{j} - 3\hat{k}$ should be applied to produce a vector moment $\vec{M} = 3\hat{i} - 2\hat{j} + \hat{k}$ about $(-1, 2, -3)$.

Example 25: Find the volume of a parallelopiped with adjacent sides $\vec{a} = \hat{i} - 2\hat{j} + \hat{k}$,

$$\vec{b} = 3\hat{i} - 3\hat{j} + \hat{k} \text{ and } \vec{c} = -\hat{i} - \hat{j} + 2\hat{k}.$$

Example 26: Find the volume of a tetrahedron with vertices $A(0, 0, 0)$, $B(1, 3, -1)$, $C(2, 2, 1)$ and $D(1, 6, 5)$.

Example 27: Find the value of λ so that vectors

$$\vec{a} = \hat{i} - \hat{j} + \hat{k}, \vec{b} = 2\hat{i} + \hat{j} + \hat{k} \text{ and } \vec{c} = -\hat{i} + \lambda\hat{j} + 2\hat{k} \text{ are coplanar.}$$

4. i. Find the value of λ if the points $A(-1, 4, -3)$; $B(3, \lambda, -5)$; $C(-3, 8, -5)$ and $D(-3, 2, 1)$ are coplanar.

ii. If the vectors $\vec{a} = \alpha\hat{i} + \hat{j} + \hat{k}$; $\vec{b} = \hat{i} + \beta\hat{j} + \hat{k}$ and $\vec{c} = \hat{i} + \hat{j} + \gamma\hat{k}$ are coplanar, then prove that $\frac{1}{1-\alpha} + \frac{1}{1-\beta} + \frac{1}{1-\gamma} = 1$ where $\alpha, \beta, \gamma \neq 1$

Chapter # 04

Example 5: The third term of an arithmetic sequence is 8, and the sixteenth term is 47. Find a_1 , d and construct the sequence. Also find a_{15} .

Example 6: Find the four arithmetic means between 19 and 54.

Example 8: Find the 7 *A.Ms* between 7 and 20.

15. Find x and y if 2 and 13 are two arithmetic means between x and y .

9. If $\frac{1}{a}$, b , $\frac{1}{c}$ are in A.P. Show that the common difference is $\frac{a-c}{2ac}$.

6. The fifth term of an arithmetic sequence is 19 and 11th term is 43. Find the first term and 87th term.

7. Which term of the sequence $-6, -2, 2, \dots$ is 70? 8. Which term of the sequence $\frac{5}{2}, \frac{3}{2}, \frac{1}{2}$ is $-\frac{105}{2}$?

Example 11: Theaters are often built with more seats per row as the rows move towards the back. Suppose the main floor of a theater has 28 seats in the first row, 32 in the second, 36 in the third and so on for 50 rows. How many seats are on the main floor?

Example 12:

Find the first three terms of an arithmetic series where $a_1 = 17$, $a_n = 101$ and $S_n = 472$.

23. A formation of a marching band has 14 marchers in the front row, 16 in the second row, 18 in the third row and so on, for 25 rows. How many marchers are in the last row? How many marchers are there altogether?

24. How many poles will be in a pile of telephone poles if there are 50 in the first layer, 49 in the second and so on, until there are 6 in the last layer?

Example 16:

Mr. Khalid saves Rs. 1000 on the first day. Then each day thereafter, saves double the amount he saved the day before. Find the amount he should save the 20th day of the month.

Example 18: Find two geometric means between 81 and 3.

Example 19: A vacuum pump removes $\frac{1}{5}$ of the air from a sealed container on each stroke of its piston. What percent of the air remains after five stroke of the piston?

26. A Ping-Pong ball is dropped from a height of 16 ft and always rebounds one-fourth of the distance fallen. How high does it rebound the 6th time?
27. A city has a current population of 100,000 and the population is increasing by 3% each year. What will the population be in 15th years?

Example 21: Find the sum of a geometric series for which $a_1 = 48$, $a_n = 3$ and $r = -\frac{1}{2}$.

Example 22: Find a_1 in a geometric series where $S_7 = 3279$ and $r = 3$.

Example 23: Find the total distance travelled by the pendulum before coming to rest, if its successive swings form the geometric series:

$$25 + 20 + 16 + 12.8 + \dots$$

Example 24: Find the sum of the infinite geometric series $\frac{4}{3} - \frac{2}{3} + \frac{1}{3} - \frac{1}{6} + \dots$

Example 25: Find fractional notation for 0.63636363...

16. A hot-air balloon rises 80 ft in the first minute of flight. If in each succeeding minute the balloon rises only 90% as far as in the previous minute, what will be its maximum altitude if it is allowed to rise without limit?

Example 27: Find the harmonic sequence, whose fourth term is $\frac{1}{13}$ and eleventh term is $\frac{1}{25}$.

8. 7, 4, 1, . . . is arithmetic sequence, find the 17th term of H.P.

Example 31: Write sigma notation for the sum.

- a) $1 + 4 + 9 + 16 + 25$
 b) $-1 + 3 - 5 + 7$
 c) $3 + 9 + 27 + 81 + \dots$

Example 34: Find the sum of $1 + \frac{3}{2} + \frac{5}{4} + \frac{7}{8} + \dots$ to n terms.

Example 35: Find the sum to infinity of the arithmetic-geometric series:

$$1 + \frac{2}{3} + \frac{3}{9} + \frac{4}{27} + \frac{5}{81} + \dots$$

Sum to n terms of the following series (arithmetico-geometric series):

$$25. \quad 1 + \frac{4}{7} + \frac{7}{7^2} + \frac{10}{7^3} + \dots$$

$$26. \quad 1 + \frac{7}{2} + \frac{13}{4} + \frac{19}{8} + \frac{25}{16} + \dots$$

Find sum to infinity of the following series:

$$27. \quad 5 + \frac{7}{3} + \frac{9}{9} + \frac{11}{27} + \dots$$

$$28. \quad 1 + \frac{2}{5} + \frac{3}{25} + \frac{4}{125} + \dots$$

$$29. \quad 1 + 4x + 7x^2 + 10x^3 + \dots$$

Example 37: A new virus is on a remote area. On day one, there were 10 people infected, with the number of new infections increasing at a rate of 40% per day.

- Find the expected number of infected people on the 7th day.
- Find the expected number of infected people during week (7 days).

Example 36: Khalid is saving for a new car. He deposits Rs. 100,000 into his account and then each month he deposits in Rs. 10,000 more than the month before. If the price of the car is Rs. 1,260,000; find:

- The amount Khalid has saved in four months.
- The time in which Khalid reaches his goal of Rs. 1,260,000.

Chapter # 05

Example 3: Find the value of p , when $3x^4 - 4px^2 + 5x - p$ is divided by $x + 2$ and the remainder is 4.

Example 6: If one zero of $g(x) = 2x^3 + x^2 - 2x - 1$ is $-\frac{1}{2}$, find its other zeros.

Example 8: Determine the value of k for which $x + 3$ is a factor of $(x + 2)^5 + (3x + k)$.

5. Find the value of ' q ' if $x^3 + qx^2 - 7x + 6$ is exactly divisible by $(x + 1)$.
6. Find the value of ' m ' in the polynomial $2x^3 + 3x^2 - 3x - m$ which when divided by $x - 2$ gives the remainder of 16.
6. If $(x - 2)$ is one of the factor of $2x^3 - 15x^2 + 16x + 12$, find its other factors.

7. Factorize $2x^3 - 15x^2 + 27x - 10$ if ' $\frac{1}{2}$ ' is one of its zero.

8. If $h(x) = 4x^3 + 4x^2 + 73x + 36$ and $h\left(\frac{-1}{2}\right) = 0$, then factorize $h(x)$.

Example 11: The volume of a rectangular solid is 72 cubic units. The width is twice the height and the length is 7 units more than the height. Find the dimensions of the solid.

6. The volume of the box is $y^3 - 2y^2 - y + 2$. If the length of one side is $y - 2$, find the length of the other two sides.
3. A rectangular solid has a volume of 144 cubic units. The width is twice the height and the length is 2 units more than the width. Find the dimensions of the solid.

Chapter # 06

Example 8: There are 5 men and 5 women in a party. Find the number of ways in which they can be seated at a round table if:

- (i) anyone can occupy any seat, (ii) men and women have alternate seats.

Example 6: How many different arrangements of the letter used in the word EVENING can be made by using all the letters?

Prove the following for $n \in \mathbb{N}$.

(iv). ${}^nP_r = {}^{n-1}P_r + r \cdot {}^{n-1}P_{r-1}$ 2. Find n , if:

(ix). ${}^{2n-1}P_n : {}^{2n+1}P_{n-1} = 22 : 7$

(viii). ${}^{n-1}P_3 : {}^{n+1}P_3 = 5 : 12$

(iii). $16 {}^nP_3 = 13 {}^{n+1}P_3$ 3. Find r , if:

(vi). $2^6 P_{r-1} = {}^5P_r$ (iv). ${}^{10}P_r = 3 {}^{10}P_{r-1}$

16. How many odd numbers can be formed by using the digits 1, 2, 3, 4, 5, 6 when repetition of digits is not allowed.
17. How many 4-digit odd numbers can be formed using the digits 1, 2, 3, 4 and 5 if no digit is repeated.
18. How many odd numbers less than 10,000 can be formed using the digits 0, 2, 3, 5, 6 without repeating the digits.
4. How many 3-digit even numbers can be formed from the digits 1, 2, 3, 4, 5, 6, if the digits are not repeated?
5. How many 7-digits mobile number can be made using the digits 0 to 9, if each number starts with 5 and no digit is repeated?
6. How many 4-digit numbers can be formed with the digits 1, 2, 3, 4, 5, 6 when the repetition of the digits is allowed?

12. Find the number of arrangement of letters of the word VOWEL in which vowels may occupy odd places?
13. In how many ways can letters of word MACHINE be arranged so that all the vowels are never together?

Example 9: 10 students applied for 6 HEC scholarships. In how many ways can these 6 be chosen?

2. Find n , if: (vi) ${}^{2n}C_3 : {}^nC_2 = 12 : 1$

(v) ${}^nC_6 : {}^{n-3}C_3 = 33 : 4$ 3. Find r , if:

(iv) ${}^{15}C_r : {}^{15}C_{r-1} = 11 : 5$

4. Find n and r , if:

(ii) ${}^nC_{r-1} : {}^nC_r : {}^nC_{r+1} = 3 : 4 : 5$

8. There are 10 points on a circle. Find the number of (i) lines (ii) triangles that can be drawn?
9. Find the number of diagonals in n sided polygon?
10. In how many ways a group of 10 girls can be divided into two groups of 3 and 7 girls.
11. Number of diagonals in n -sided polygon is 35. Find the number n ?
13. In an examination, a candidate has to pass in each of 6 subjects. In how many ways he cannot qualify the examination?
6. Out of 5 men and 3 women, a committee of 3 is to be formed. In how many ways can it be formed if at least one man is selected?

Chapter # 07

Example 1: Use the method of mathematical induction to prove that

$$1.2 + 2.3 + 3.4 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$$

for all positive integers ' n '.

Example 2: Use the method of mathematical induction to prove that

$$\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right) \dots \left(1 - \frac{1}{n+1}\right) = \frac{1}{n+1}$$

for all positive integers ' n '.

Example 3: Use the method of mathematical induction to show that $n^2 - 3n + 4$ is an even (i.e.; divisible by 2) for all positive integers n .

Example 4: Use the method of mathematical induction to show that $3^n > n^2$ for all positive integers n .

Example 5: Use the method of mathematical induction to show that

$$4 + 4.6 + 4.6^2 + 4.6^3 + \dots + 4.6^n = \frac{4(6^{n+1}-1)}{5} \text{ for all positive integers } n.$$

By the method of the mathematical induction prove the following when n is an integer.

$$4. \quad \frac{1}{1.4} + \frac{1}{4.7} + \frac{1}{7.10} + \dots + \frac{1}{(3n-2)(3n+1)} = \frac{n}{3n+1} \quad \forall n \geq 1$$

$$19. \quad \binom{n}{0} + \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} + \dots + \frac{1}{n+1}\binom{n}{n} = \frac{2^{n+1}-1}{n+1} \quad \forall n \geq 0$$

$$23. \quad 4^n + 15n - 1 \text{ is divisible by } 9 \quad \forall n \geq 1$$

$$24. \quad 7^n - 2^n \text{ is divisible by } 5 \quad \forall n \geq 0$$

$$26. \quad 5^n + 9 < 6^n \quad \forall n \geq 2$$

$$31. n^3 > 2n + 1$$

$$\forall n \geq 2$$

$$32. n! > n^2$$

$$\forall n \geq 4$$

Example 9: Find the constant term in the expansion of $\left(x + \frac{2}{x}\right)^{10}$.

Example 10: Find the 3rd term from the end in the expansion of $\left(2 - \frac{5}{\sqrt{x}}\right)^5$.

Example 11:

Find the remainder when 7^{101} is divided by 25.

Example: The fourth term in the expansion of $\left(ax + \frac{1}{x}\right)^n$ is $\frac{5}{2}$

Find the values of a and n .

5. Find the middle term in the expansion of the following.

$$(i) \left(3x^2 - \frac{1}{2x}\right)^{10}$$

$$(ii) \left(2x^2 - \frac{1}{5x}\right)^{11}$$

$$(iii) \left(\frac{a}{\sqrt{x}} + \sqrt{x}\right)^8$$

$$(iv) \left(a - \frac{3}{x^2}\right)^{12}$$

6. Find the specified term in the following expansions.

(ii) Term involving q^8 in the expansion of $\left(\frac{p^2}{2} + 6q^2\right)^{12}$

7. Find the term independent of x in the expansions of the following.

$$(i) \left(2x^2 - \frac{1}{x}\right)^{12}$$

$$(ii) \left(\sqrt{x} + \frac{1}{3x^2}\right)^{10}$$

12. If the coefficients of 2nd, 3rd and 4th terms in the expansion of $(1+x)^{2m}$ are in A.P., show that $2m^2 - 9m + 7 = 0$.

4. If x is so small that its square and higher powers may be neglected then prove that:

$$(iii) \frac{\sqrt{9-x} + \left(1 + \frac{3}{4}x\right)^{-5}}{2+x} \approx 2 - \frac{71}{24}x$$

8. Identify as binomial series and find the sum of the following:

$$(iv) \quad 3 - \frac{3}{18} - \frac{3}{2!.18^2} - \frac{3^2}{3!.18^3} - \dots$$

9. (i) If $y = \frac{3}{2^2.1!} + \frac{3}{2^4.2!} + \frac{3}{2^6.3!} + \dots$, then show that $8y^2 + 16y - 19 = 0$.

(ii) If $y = \frac{1}{3} + \frac{1.3}{3.6} + \frac{1.3.5}{3.6.9} + \dots$ then show that $y^2 - 2y - 2 = 0$.

Example 13: Find the unit digit of: (i) 17^{203} (ii) 29^{26} (iii) 36^{307}

Example 14: Find the remainder when 7^{103} is divided by 25.

Example 15: If the fractional part of the number $\frac{2^{403}}{15}$ is $\frac{k}{15}$, then find k .

Example 16: Find the unit digit of (i) 17^{203} (ii) 29^{26} (iii) 36^{307}

Example 18: Which one is greater $99^{50} + 100^{50}$ or 101^{50} ?

Example 19: Show that $11^9 + 9^{11}$ is divisible by 10.

Example 17: Find the last two digits of the number $(13)^{10}$.

13. Show that $6^{n+3} - 8n - 6$ is divisible by 6.

4. Find the last one or two digits of the number where applicable.

a. 15^8

b. 37^7

c. 29^{10}

6. Show that $12^{15} + 8^{15}$ is divisible by 10.
9. What is the remainder when 17^{1717} is divided by 9.
8. Use binomial theorem to find the remainder when 5^{103} is divided by 13.

Chapter # 08

Example 3: If $\sin A = \frac{1}{5}$, and $\sin B = \frac{1}{3}$, where angles A and B are in quadrant II, then find $\cos(A + B)$. In which quadrant does the terminal arm of angle $(A + B)$ lie?

Example 5:

Express $12\sin\theta + 5\cos\theta$ in the form of $r \sin(\theta + \phi)$.

5. For $\sin\alpha = \frac{4}{5}$, $\tan\beta = -\frac{5}{12}$ with terminal sides of angles α and β lie quad. II, find $\cos(\alpha + \beta)$ and $\cos(\alpha - \beta)$.

Example 4: If α, β, γ are interior angles of a triangle, then prove that:

$$\cot\beta \cot\gamma + \cot\alpha \cot\gamma + \cot\alpha \cot\beta = 1$$

8. If $\sin\alpha = \frac{3}{5}$, where $0 < \alpha < \frac{\pi}{2}$ and $\cos\beta = \frac{12}{13}$, where $\frac{3\pi}{2} < \beta < 2\pi$ find:

(i) $\csc(\alpha + \beta)$ (ii) $\sec(\alpha + \beta)$ (iii) $\cot(\alpha + \beta)$

9. Given α and β are obtuse angles with $\sin\alpha = \frac{1}{\sqrt{2}}$ and $\cos\beta = -\frac{3}{5}$ find:

(i) $\sin(\alpha \pm \beta)$ (ii) $\cos(\alpha \pm \beta)$ (iii) $\tan(\alpha \pm \beta)$

12. If $\alpha + \beta + \gamma = 180^\circ$, prove that:

(i) $\tan\alpha + \tan\beta + \tan\gamma = \tan\alpha \tan\beta \tan\gamma$ (ii) $\cot\frac{\alpha}{2} + \cot\frac{\beta}{2} + \cot\frac{\gamma}{2} = \cot\frac{\alpha}{2} \cot\frac{\beta}{2} \cot\frac{\gamma}{2}$

(iii) $\tan\frac{\alpha}{2} \tan\frac{\beta}{2} + \tan\frac{\beta}{2} \tan\frac{\gamma}{2} + \tan\frac{\gamma}{2} \tan\frac{\alpha}{2} - 1 = 0$

11. Show that:

$$\begin{aligned} \text{(v)} \quad \frac{\tan(x+y)}{\cot(x-y)} &= \frac{\tan^2 x - \tan^2 y}{1 - \tan^2 x \tan^2 y} & \text{(vi)} \quad \frac{\cos(\alpha + \beta)}{\cos(\alpha - \beta)} &= \frac{1 - \tan \alpha \tan \beta}{1 + \tan \alpha \tan \beta} \\ \text{(vii)} \quad \cot(\alpha - \beta) &= \frac{\cot \alpha \cot \beta + 1}{\cot \beta - \cot \alpha} & \text{(viii)} \quad \frac{\cos 4\theta}{\csc \theta} + \frac{\sin 4\theta}{\sec \theta} &= \sin 5\theta \end{aligned}$$

Example 7: Use the half-angle identities to find exact values for: (a) $\sin 15^\circ$ (b) $\tan 15^\circ$

Example 8: For $\cos \alpha = \frac{-7}{25}$ and α in quad. III, find values of $\sin\left(\frac{\alpha}{2}\right)$ and $\cos\left(\frac{\alpha}{2}\right)$.

Example 9: Express $4\sin^4 x$ in terms of an expression containing only cosines to the power 1.

Example 10: Find the exact value of $\sin 22.5^\circ \cdot \cos 22.5^\circ$.

Example 11: Prove that: $\tan\left(\frac{\theta}{2}\right) = \pm \left(\frac{1 - \cos \theta}{\sin \theta}\right)$

8. Verify the following identities.

$$\text{(ix)} \quad \cos 4\theta = 8\cos^4 \theta - 8\cos^2 \theta + 1$$

$$\text{(iii)} \quad \tan \frac{\theta}{2} = \frac{\sin \theta}{1 + \cos \theta}$$

$$\text{(v)} \quad 8\sin^4 \theta = 3 + \cos 4\theta - 4\cos 2\theta$$

$$\text{(xv)} \quad \frac{\sin 3\alpha}{\sin \alpha} - \frac{\cos 3\alpha}{\cos \alpha} = 2$$

$$(xvii) \quad \frac{\sin \theta}{1 + \cos \theta} = \tan \frac{\theta}{2}$$

$$(xix) \quad \frac{\sin 2\alpha}{\sin \alpha} - \frac{\cos 2\alpha}{\cos \alpha} = \sec \alpha$$

Example 14: Simplify $\sin 40^\circ \cos 20^\circ + \cos 40^\circ \sin 20^\circ$

Example 17: Show that: $\cos^2 6\alpha + \cos^2 5\alpha + \cos^2 3\alpha + \cos^2 2\alpha = 4 \cos(4\alpha) \cos(1.5\alpha) \cos(0.5\alpha)$

Example 18: Show that: $\sin 70^\circ \sin 30^\circ \cos 20^\circ \cos 10^\circ = \frac{1}{4} \cos 10^\circ + \frac{\sqrt{3}}{16} + \frac{1}{8} \sin 40^\circ$

Solution: $\sin 70^\circ \sin 30^\circ \cos 20^\circ \cos 10^\circ$

4. Prove that.

$$(i) \quad \cos 80^\circ \cos 60^\circ \cos 40^\circ \cos 20^\circ = \frac{1}{16} \quad (ii) \quad \sin 70^\circ \sin 50^\circ \sin 30^\circ \sin 10^\circ = \frac{1}{16}$$

$$(iii) \quad \sin \frac{\pi}{9} \sin \frac{2\pi}{9} \sin \frac{3\pi}{9} \sin \frac{4\pi}{9} = \frac{3}{16}$$

3. Prove the following identities.

$$(iv) \quad \sin 3\theta + \sin \theta = 4 \cos^2 \theta \sin \theta$$

$$(viii) \quad \frac{\cos 3\theta + \cos \theta}{\cos 3\theta - \cos \theta} = -\cot 2\theta \cot \theta$$

$$(ix) \quad \frac{\cos 6x + \cos 8x}{\sin 6x - \sin 4x} = \cot x \cos 7x \sec 5x$$

Chapter # 09

Example 1:

Find domain and range of the following:

(i) $y = 4 \sin 3x$

(ii) $y = \frac{1}{2 \cos x - 1}$

(iii) $y = \frac{1}{1 + 2 \sin \theta}$

(iv) $y = \frac{1}{2 - \sin 3x}$

Example 3: Find the period of $f(x) = \cot 3x + \sin \frac{2x}{3}$.

Example 4: Find the period of $f(x) = 7 \sin (3x + 5)$.

Example 5: Find the maximum and minimum values of $y = 3 + 4 \sin \theta$.

Example 6: Find the maximum and minimum values of $y = \frac{1}{3} - 5 \cos \theta$.

Example 7: Find the maximum and minimum values of the following trigonometric functions.

(i) $y = 1 + 2 \sin \theta$ (ii) $y = 3 + 2 \cos (3\theta - 2)$ (iii) $y = \frac{1}{1 + 3 \sin (2\theta - 15)}$

2. Find the maximum and minimum values of the following reciprocal trigonometric functions.

(i) $y = \frac{1}{4 + 3 \sin \theta}$

(ii) $y = \frac{1}{\frac{1}{2} - 5 \cos \theta}$

(iii) $y = \frac{1}{\frac{1}{3} - 4 \sin (2\theta - 5)}$

(iv) $y = \frac{1}{3 + \frac{2}{5} \sin (5\theta - 7)}$

1. Find the maximum and minimum values of the following trigonometric functions.

(iii) $y = \frac{1}{5} - 2 \sin (3\theta - 7)$

(iv) $y = 7 + \frac{3}{5} \cos (2\theta - 1)$

3. Find domain and range of the following:

(iv) $y = 7 \cot \frac{\pi}{2} x$

(vi) $y = \operatorname{Cosec} 4x$

(iii) $y = \sin \frac{2x}{3}$

(v) $y = 4 \tan \pi x$

4. Find the periods of the following:

(i) $y = 6 \sec (2x - 3)$ (ii) $y = \cos (5x + 4)$ (iii) $y = \cot 4x + \sin \frac{5x}{2}$

(iv) $y = 7 \sin (3x + 3)$ (v) $y = 5 \sin (2x + 3)$ (vi) $y = 2 \tan 3x + 7 \cos 5x$

Example 9:

Check whether the following function is odd or even?

(i) $y = 3 \sin x + 4 \cos x$ (ii) $y = \frac{\tan x}{x - \sin x}$

Example 11:

The most used current is alternating current and it reverses direction in a cyclic fashion. In Canada the standard frequency is 60 Hz (60 cycles per second). The maximum voltage is about 170 V. the voltage can be modeled by $V(t) = a \sin[K(t - d)] + c$.

1. Check whether the following functions are odd or even?

(v) $y = \frac{\sin^2 x}{x + \tan x}$

(vi) $y = \frac{\tan x - \sin x}{\sin^3 x}$

2. Draw the graph of each of the following functions:

(ii) $y = 2 \cos 3x$

(iv) $y = \cos \frac{x}{2}$

(x) $y = \sin \frac{x}{2}$

6. Alternating current cyclically reverses direction. The maximum voltage is about 180 volts when the standard frequency is 56 Hz (56 cycles per second). The voltage can be modeled by

$$V(t) = a \sin(k(t - d)) + c$$

Determine each of the following:

- Period of 56 Hz AC,
 - The value of k ,
 - The amplitude of the voltage function,
 - Model the voltage with an appropriate transformed Sine function.
8. The pendulum of a clock swings with a periodic motion that can be represented by a trigonometric function. At rest, the pendulum is 16 cm above the base. The highest point of the swing is 20 cm above the base, and it takes 2 seconds for the pendulum to swing back and forth to its starting point once. Assume the pendulum is released from its highest point. Write a cosine equation that models the height of the pendulum as a function of time.